

① Suppose that $DS = 0250H$, $BX = 0370H$, and $DI = 400H$. Determine the memory address accessed by each of the following instructions assuming real mode operation:

a) $MOV AL, [3408H]$

Given, $DS = 0250H$

$BX = 0370H$

$DI = 0400H$

Computing the base address of DS (Data Segment)

$$\begin{aligned} DS \times 10H &= 0250H \times 10H \\ &= 02500H \end{aligned}$$

We know, in real mode,
the physical memory address is calculate as:

$$\text{Physical Address} = (\text{Segment} \times 10H) + \text{Offset}$$

Here, $\text{Offset} = 3408H$

$\text{Segment} = 02500H$

$$\begin{aligned} \text{So, Physical Address} &= (0250H \times 10H) + 3408H \\ &= \boxed{05908H} \end{aligned}$$

b) MOV [DI], AL

Here, this instruction uses register indirect addressing using DI.

$$DI = 0400H$$

$$DS = 0250H$$

$$\begin{aligned}\therefore \text{Physical Address} &= (0250H \times 10H) + 0400H \\ &= \boxed{02900H}\end{aligned}$$

c) MOV DL, [BX + DI]

Here, this instruction uses based-indexed addressing.

$$BX = 0370H$$

$$DI = 0400H$$

$$\begin{aligned}\therefore \text{Effective Address (Offset)} &= 0370H + 0400H \\ &= 0770H\end{aligned}$$

$$\begin{aligned}\therefore \text{Physical Address} &= (0250H \times 10H) + 0770H \\ &= 02500H + 0770H \\ &= \boxed{02C70H}\end{aligned}$$

② In real mode, show the starting and ending addresses of each segment located by the following segment register values:

a) 1000H

In 8086 Real Mode:

$$\text{Physical Address} = (\text{Segment} \times 10\text{H}) + \text{Offset}$$

Since a segment is 64KB long,

$$\text{Starting address} = \text{Segment} \times 10\text{H} + 0000\text{H}$$

$$\text{Ending address} = \text{Segment} \times 10\text{H} + \text{FFFFH}$$

$$\text{So, Starting address} = 1000 \times 10\text{H} = \boxed{10000\text{H}}$$

$$\text{Ending address} = 10000\text{H} + \text{FFFFH} = \boxed{1\text{FFFFH}}$$

b) 1234H

$$\text{Starting address} = 1234 \times 10\text{H} = \boxed{12340\text{H}}$$

$$\begin{aligned} \text{Ending address} &= 12340\text{H} + \text{FFFFH} \\ &= \boxed{2233\text{FH}} \end{aligned}$$

③ Convert the following:

a) MOV BX, SS:[BP+DI] from assembly language to machine language.

Given assembly code,

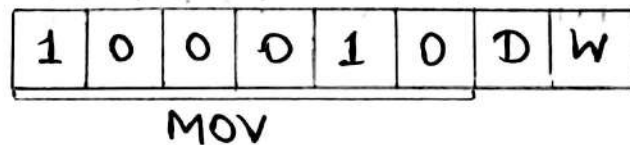
MOV BX, SS:[BP+DI]

Where, Destination = BX (16bit register)

Source = [BP+DI] (16bit memory operand)

Therefore, it matches the "MOV r16, r/m16" instruction.

The instruction:



Direction bit = 1 [Memory to Register]

Word bit = 1 [16bit operand]

Therefore, $(10001011)_2$

↓ ↓
B B

For REG field

BX = 011

For R/M field

[BP+DI] = 011

For MOD field

MOD = 00

MOD	REG	R/M
00	011	011

Therefore, $(00011011)_2$

↓ ↓
1 B

Since the instruction explicitly specifies SS, the default segment is: SS

Now, first byte = 8B

second byte = 1B

∴ Machine code: 8B 1B

b) 8B07H from machine language to assembly language.

Here, first byte = 8BH

Therefore, instruction = MOV r16, r/m16

operation = MOV

Then, second byte = 07H

$(07)_{16} \rightarrow (00000111)_2$

MOD		REG			R/M		
0	0	0	0	0	1	1	1

∴ REG = 000

Destination Register = AX

∴ R/M = 111

Addressing Mode = [BX]

∴ MOD = 00

There will be no displacement.

∴ Assembly instruction: MOV AX, [BX]

④ Suppose, Core2 microprocessor operates in protected mode into Windows OS. To perform read/write operation, every microprocessor needs to calculate the physical address of RAM/ROM in protected mode. The segment register & descriptor table are used for this purpose. Consider the following:

i) Define the address bit of the descriptor so that the microprocessors use up to 4GB of RAM.

Maximum addressable size in protected mode:

$$\text{Limit} = \text{FFFFFFH} \quad (20 \text{ bits})$$

When $G=1$, the limit is interpreted in 4KB units.

Verification:

$$\begin{aligned} & (\text{Limit} + G) \times 4 \text{ KB} \\ &= (\text{FFFFFF} + 1) \times 4 \text{ KB} \\ &= 100000 \text{ H} \times 1000 \text{ H} \\ &= 4 \text{ GB} \end{aligned}$$

So,

$$\text{Limit} = \text{FFFFFFH}$$

$$G = 1$$

ii) You can use 64KB (or max 16MB) for any segment of RAM

Since the Core2 works in 32-bit mode,

$$D/B = 1 \quad (32\text{-bit segment})$$

⑥

iii) You must define the descriptor privilege level of access right portion so that the segment access will be granted.

Assuming,

DPL	Privilege
0	Kernel
1	OS
2	Driver
3	User

Normal windows apps will run at the "User Level".

Therefore, $DPL = 3 = (11)_2$

iv) Set a bit of access right portion that ensures CS (code segment) has a valid address and limit.

The Present (P) bit defines whether the segment available or not.

Therefore, Present (P) = 1

v) Ensure the descriptor describes code segment portion.

Code Segment (S) = 1

Executable bit (E) = 1

vi) Confirm that the code segment works through privilege level.

The code segment operates on the "User" Level
Since $DPL = (11)_2$

vii) Define a bit in the access right portion that ensures if necessary code segments may be read.

Read (R) bit defines,

$R = 0$ [Execute only]

$R = 1$ [Execute & Read]

$\therefore$ We set $R = 1$ to make the code segment readable.

viii) Define a bit in the descriptor so that segment extension feature will be accepted.

Conforming (C) bit defines,

$C = 0$ [Non-conforming]

$C = 1$ [Extension allowed]

$\therefore$ We set $C = 1$.

ix) Ensure 32-bit instruction mode

D/B bit = 1

x) Confirm 32-bit compatibility mode.

Long-mode (L) bit,

$L = 0$ [32-bit compatibility]

$L = 1$ [64-bit code]

$\therefore L = 0$

xi) Ensure code segmentation is available.

Present (P) bit = 1

Therefore,

$P = 1$

$DPL = 11$

$S = 1$

$E = 1$

$C = 1$

$R = 1$

$A = 0$

P	DPL	S	E	C	R	A
1	1	1	1	1	1	0

$\rightarrow (11111110)_2$

$\rightarrow (FE)_h$

$\therefore$ Access byte = FEH

Flags :

G	D/B	L	AVL
1	1	0	0

$\rightarrow (1100)_2$

$\rightarrow CH$

$\therefore$ Flags byte = CH

Now Base = 00000000H

Limit = FFFFFFFH

Flags + Limit (19:16) = F

Descriptor layout :

Base (31:24)	00
Flags + Limit 19:16	CF
Access byte	FE
Base 23:16	00
Base 15:0	0000
Limit 15:0	FFFF

∴ The complete descriptor becomes :

00 CFFE 000000 FFFF H